

The Fine Tuning Argument

- The Universe is dependent on very exact values of constants
- If they were changed even a little bit, life (or the Universe itself) wouldn't exist
- "a very fine tuning" is required to support life!

The Argument

- The chance of all these parameters being exactly right is extremely unlikely by chance
- It is more likely that a designer intentionally set these parameters

O = Constants: Physical constants support life.

H1 = Design: Constants set by creator

H2 = Naturalism: Brute chance of constants being this way

- Can use likelihood argument to support Design

Counterarguments:

- It's possible we could still support life with many different constants; we just haven't tested them yet.
- Independent evidence issue (see biological design argument)
- Designer may not have wanted/ been able to fine tune
- The Multiverse Argument
 - if there are a large/infinite number of universes, it is very likely that by random chance, some will support life
 - most analogous to natural selection for biological design
 - Problem: Regardless of if there is one or infinite universes, the probability that life exists is statistically independent. The probability that our universe supports life is therefore the same under single- or multi-verse theories.
 - ↳ Inverse Gambler's Fallacy: A person observes a double-6 being rolled.
 - Observation: this particular roll is double-6.
 - doesn't matter if they have been rolling all day (independent)
 - Observation: A double-6 was thrown.
 - highly likely they have been rolling all day
 - Principle of Total Evidence: must use all evidence we are given and not fall back to weaker observations.

Anthropic Principle

If there exists a universe unsuitable for life, we would not be observe it.

Our existence explains why the physical constants are the way we are. (Strong Anthropic Principle)

Observational Selection Effect: We must conditionalize probabilities on the methods we use to make observations.

Example: if you catch fish with a net and all are >10 in, you might conclude that the evidence supports all fish being >10 in. However, if the net has 10-inch holes, the evidence means nothing.

↳ If O = all caught fish >10 in:

$P(\text{fish} > 10 \text{ in} | \text{net with } 10 \text{ inch holes}) =$

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all fish being >10in. However, if the net has 10-inch holes, the evidence means nothing.

↳ If $O =$ all caught fish >10 in:

$$P(\text{All} > 10 | O) > P(\text{half} > 10 | O)$$

However,

$$P(\text{All} > 10 | O \& \text{net w/ holes}) = P(\text{half} > 10 | O \& \text{net w/ holes}) = 1.$$

Weak Anthropic Principle:
true, but irrelevant

$$Y \text{ implies } X, \text{ so } P(X|Y) = 1$$

I can add another aux. hypothesis, e.g. "I exist" via Weak Anthropic Principle, but since it will not affect the observation it is still valid

The OSE applies to the fine tuning argument:

$$P(\text{Support Life} | \text{Naturalism} \& \text{We Exist}) = P(\text{Support Life} | \text{Design} \& \text{We Exist}) = 1.$$

↳ Conclusion: There is no evidence of either fine tuning or naturalism.

(Weak Anthropic Principle)

"We exist" should always be an auxiliary hypothesis

"What we can expect to observe must be restricted by our role as observers"

↓

Revised Likelihood Principle: O supports X more than Y iff

$$P(O|X \& M) > P(O|Y \& M) \text{ where } M \text{ is the method used to collect observations.}$$

↳ different from weak anthropic principle: simply suggests we should take into account the methods we use

The Firing Squad Objection ← What counts as a "method"?

• You are sentenced to death by firing squad.

• All the shooters miss.

Two Hypotheses:

#1 • There was a conspiracy to save you.

#2 • All shooters missed by random chance.

↳ Intuition: #1 is more likely than #2.

↳ Anthropic Principle: $P(O | \#1 \& \text{Survived}) = P(O | \#2 \& \text{Survived}) = 1$. There is no evidence of either hypotheses!

→ Firing Squad demonstrates a fundamental flaw in the weak anthropic principle: overgeneralizing the data collection method will result in valid hypotheses being ignored.

• Method must play a causal role in explaining how an observation came to be. In firing squad, being alive has no role in explaining why the shooters missed.

bystander argument

→ - can be refuted by looking from the perspective of a bystander:

the fact that bystander exists is irrelevant, so Anthropic principle would still hold

↳ still doesn't solve issue of observing as the person being shot